Pronunciation assessment

Overview

The goal of this project is creating a system that takes a text-audio pair of input and produces 2 outputs:

- Phoneme-level accuracy: identifies which phonemes are pronounced correctly and incorrectly

- Word stress accuracy: determines if the primary stress of the word is placed on the correct syllable

We aim to create such lightweight models that can be packaged in a reasonable small mobile app and run locally

Methodology

Preprocessing

- For text, use CMUdict to get phoneme sequence and stress pattern
  - *Example: "banana" -> B AH0 N AE1 N AH0*
  - http://www.speech.cs.cmu.edu/cgi-bin/cmudict
- For audio, convert to MFCCs for acoustic models

Phoneme-level accuracy

Step 1: Use MFA (Kaldi) to perform a forced alignment
  - Input: audio file, text transcript, CMUdict
  - Output: precise start and end timestamp fors every phoneme
  - E*xample Output: B (0.00s-0.05s), AH (0.05s-0.12s), N (0.12s-0.18s),...*

Step 2: GOP calculation
  - Input: audio and CMUdict phonemes of the word
  - Output: LLR for each phoneme (threshold this as score for displaying)
  - Example output: B (-0.29), AH (-0.12),...

=> Use [dataset_L2ARCTIC] and [dataset_speechocean762] for thresholding

[method_1.pdf] - classic GOP algorithm using LLR from GMM-HMM systems
[method_2.pdf] - modern DNN-based GOP

Word stress accuracy [cao24b.pdf]

Step 1: Use the phonemes timestamps from the above-mentioned, group them into syllables. Rule based: **Maximal Onset Principle**

Step 2: Use wav2vec to extract feature vecs for each syllable. Use padding to create fixed-length sequences

Step 3: Train an Attention model, mark the most condifent syllable as stressed